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Sterilization container and latch

Abstract

A sterilization container for medical instruments comprising a base and a lid. A latch may be attached to the lid. The latch may include a mount body and a lever body pivotably coupled to the mount body such that the lever body and the mount body cooperate to define an interior having an opening. A deflector may be positioned within the interior and movable with the lever body. The deflector may include a hub portion supported, a wing portion extending from the hub portion to positioned adjacent to the opening of the interior such that when the lever body is in a secured position the wing portion prevents access to the interior through the opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The subject patent application claims priority to, and all the benefits of, United States Provisional Patent Application No. 63/334,272, filed on Apr. 25, 2022, the entire contents of which are incorporated by reference herein.

BACKGROUND

(1) Prior to a medical or surgical procedure, the tools and instruments to be utilized during the procedure are sterilized in order to avoid the spread of contaminants. In some cases, the instruments are sterilized inside a durable and reusable sterilization container, which is then used to transport the sterilized instruments from the sterilizing equipment to a treatment location, such as an operating room. Transporting instruments in this manner allows their sterility to be maintained until the moment the tool is removed from the container. Because the exterior of the sterilization container is non-sterile following transportation to the treatment location, it is desirable to minimize contact with the container when the container is opened.

SUMMARY

(2) In one aspect, a sterilization container for medical instruments. The sterilization container may comprise a base and a lid configured for engaging the base. The base and the lid may collectively define a volume for receiving medical instruments. The sterilization container may further comprise a latch, which may be attached to one of the base and the lid. The latch may comprise a mount body including a back portion and two side portions, wherein the back portion is coupled to the lid or the base. The latch may further comprise a lever body including two side portions. The two side portions of the lever body may be pivotably coupled to the two side portions of the mount body such that the lever body is movable about a pivot axis between a secured position and an unsecured position. When the lever body is pivotably coupled to the two side portions of the mount body, the lever body and the mount body cooperate to define an interior and further define an opening of the interior when the lever body is in the secured position. The latch may further comprise a deflector positioned at least partially within the interior and movable with the lever body for concurrent movement about the pivot axis. The deflector may comprise a hub portion supported on the pivot axis and a wing portion extending from the hub portion to define an outer surface. The outer surface may be positioned adjacent to the opening of the interior when the lever body is in the secured position for preventing access to the interior through the opening. The deflector may further comprise a tab coupled to the hub portion and separate from the wing portion. The tab is configured to engage the mount body when the lever body is in the unsecured position for retaining the lever body in the unsecured position.

(3) Any of the above aspects can be combined in full or in part. Any features of the above aspects can be combined in full or in part. Any of the above implementations for any aspect can be combined with any other aspect. Any of the above implementations can be combined with any other implementation whether for the same aspect or a different aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Advantages of the present disclosure will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

(2) FIG. **1** is a perspective view of sterilization container including a first implementation of a latch.

(3) FIG. **2**A is a close-up perspective view of the sterilization container and latch of FIG. **1**, with the latch shown in a secured position.

(4) FIG. **2**B is a close-up perspective view of the sterilization container and latch of FIG. **1**, with the latch shown in an unsecured position.

(5) FIG. **3**A is a side view of a user interacting with the sterilization container with the latch shown in the secured position.

(6) FIG. **3**B is a side view of a user interacting with the sterilization container with the latch shown in an intermediate position.
(7) FIG. **3**C is a side view of a user interacting with the sterilization container with the latch shown in the unsecured position.
(8) FIG. **4**A is a cross-sectional perspective view of the sterilization container and the latch of FIG. **2**A, with the latch shown in the secured position.
(9) FIG. **4**B is a cross-sectional perspective view of the sterilization container and the latch of FIG. **2**B, with the latch shown in the unsecured position.
(10) FIG. **5**A is a cross-sectional perspective view of the sterilization container and the latch of FIG. **4**A, with the latch shown in the secured position.
(11) FIG. **5**B is a cross-sectional perspective view of the sterilization container and the latch of FIG. **4**B, with the latch shown in the unsecured position.
(12) FIG. **6** is a cross-sectional perspective view of the latch of FIG. **4**A.
(13) FIG. **7** is an exploded view of the latch of FIG. **6** including a deflector.
(14) FIG. **8** is a perspective view of the deflector of FIG. **7**
(15) FIG. **9** is a side view of the deflector of FIG. **8**.
(16) FIG. **10** is a cross-sectional perspective view of a second implementation of the latch.
(17) FIG. **11** is an exploded view of the latch of FIG. **10** including a deflector.
(18) FIG. **12** is a perspective view of the deflector of FIG. **11**
(19) FIG. **13** is a side view of the deflector of FIG. **12**.
DETAILED DESCRIPTION
(20) With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a sterilization container system **100** is shown in FIG. **1**. More specifically, FIG. **1** is a perspective view of the sterilization container system **100** that includes a sterilization container **102** for sterilizing surgical and medical instruments and/or tools. In some implementations, the sterilization container system **100** may be utilized to charge a wirelessly chargeable battery. Such an implementation is described more fully in PCT Patent Application No. PCT/US2020/025429, the entire contents of which are incorporated by reference herein. In another alternative instance, the sterilization container **102** may be used to sterilize a surgical instrument other than the wirelessly chargeable batteries. For instance, the methods described herein may be used to sterilize manual surgical instruments, such as scalpels, forceps, and osteo-tomes. The methods described herein may also be used to sterilize powered surgical instruments, such as rotary handpieces, drills, or endoscopes.
(21) In one sterilization method, medical instruments may be placed within the sterilization container **102** prior to sterilization. The sterilization container **102** may then be sterilized in an autoclave process (or other suitable sterilization process) with the medical instruments contained within. Thus, in this method, the medical instruments and the sterilization container **102** may be sterilized together and a volume **120** (shown in FIGS. **4**A and **4**B) within the sterilization container **102** may be sterilized or maintained in a sterile state. The sterilization container **102** may then be carried or otherwise transported to the desired location of use while maintaining the sterility of the sterile volume **120** and medical instruments.
(22) FIG. **1** illustrates one view of the sterilization container **102**. As shown, the sterilization container **102** is substantially rectangular in shape having two sides or ends **110**, a top **112**, and a bottom **114**. However, it should be recognized that the sterilization container **102** may be any suitable shape that enables the sterilization container **102** to operate as described herein.
(23) As shown in FIG. **1**, the sterilization container **102** may comprise a lid **116** and a base **118**, which are configured for corresponding engagement with one another to collectively define a volume **120** for receiving medical instruments. Better shown in FIGS. **4**A-**5**B, the lid **116** and the base **118** are sealable to one another through use of one or more seals **122** to define the volume **120** within the sterilization container **102**. The lid **116** and the base **118** may each include a

corresponding outer surface. More specifically, the lid **116** has an outer surface **116**A and the base **118** has an outer surface **118**A. The lid **116** and the base **118** also include respective inner surfaces **116**B, **118**B, (shown in FIGS. **4** and **5**) which cooperate to define the volume **120**. In one instance, the lid **116** is removable from the base **118** to enable one or more medical instruments to be removably placed inside the sterilization container **102**.

(24) In some implementations, the lid **116** of the sterilization container **102** may include metal, which may retain heat to facilitate drying of contents thereof. For example, the sterilization container **102** may be placed in an autoclave to sterilize the contents with a high-temperature sterilant, such as steam, hydrogen peroxide, ozone, or ethylene oxide. This may result in liquid condensing on the inside of the sterilization container **102** or the contents disposed therein. After the contents are sterilized and removed from an autoclave, the lid **116** retains heat from the autoclave to facilitate drying of the contents housed within the sterilization container **102**. As such, the lid **116** may have a thermal conductivity of greater than or equal to 1 W/(m*K) at 298 Kelvin. In some instances, the lid **116** consists of, or consists essentially of, metal. In other instances, the lid **116** may not include metal. For example, the lid **116** may include a polymeric material. In such instances, the lid **116** may include a material other than metal that still facilitates drying of contents thereof by retaining heat from the autoclave.

(25) The base **118** of the sterilization container **102** includes a material having a glass transition temperature above 140° C. As previously stated, the sterilization container **102** may be placed in an autoclave to sterilize the contents with a high-temperature sterilant. As such, the base **118** includes a material having a glass transition temperature above 140° C. because temperatures inside an autoclave can exceed 120° C. The base **118** of the sterilization container **102** may also include a material permitting the transmission of electromagnetic waves therethrough. As such, the base **118** may include a material having a dielectric constant of less than or equal to ten or a dielectric constant less than or equal to five to permit the transmission of electromagnetic waves therethrough. For example, the base **118** may include a polymeric material permitting the transmission of an electromagnetic wave therethrough, such as a plastic. In another example, the base **118** may include a material other than a polymeric material that permits the transmission of electromagnetic waves therethrough, such as a glass.

(26) In one such instance, the material permitting the transmission of an electromagnetic wave therethrough may be a polymeric material and the base **118** may be formed of the polymeric material via injection molding, thermoforming, machining, 3D printing, and the like. The polymeric material may comprise the poly(aryl ether sulfone) (P) in a weight amount of at least 10%, at least 30% or at least 50%, based on the total weight of the polymeric material. Preferably, the polymeric material comprises the poly(aryl ether sulfone) (P) in a weight amount of at least 70%, based on the total weight of the polymeric material. More preferably, the polymeric material comprises the poly(aryl ether sulfone) (P) in a weight amount of at least 90%, if not at least 95%, based on the total weight of the polymeric material. Still more preferably, the polymeric material consists essentially of the poly(aryl ether sulfone) (P). The most preferably, it consists essentially of the poly(aryl ether sulfone) (P). The poly(aryl ether sulfone) (P) advantageously has a weight average molecular weight in the range of from 20,000 to 100,000. Preferably, the poly(aryl ether sulfone) (P) has a weight average molecular weight in the range of from 40,000 to 70,000. The weight average molecular weight can be determined by Gel Permeation Chromatography using conventional polystyrene calibration standards. The base **118** may comprise a polyphenylsulfone homopolymer, i.e. a polymer of which essentially (and, preferably) all the recurring units are of formula (H). RADEL® R polyphenylsulfone from SOLVAY ADVANCED POLYMERS, L.L.C. is an example of a polyphenylsulfone homopolymer.

(27) As shown throughout the figures, and particularly in FIGS. **2**A and **2**B, the sterilization container **102** may comprise a latch assembly **138**. One configuration of the latch assembly **138** is illustrated in FIGS. **2**A-**9**. Another configuration of the latch assembly **138**′ is shown in FIGS. **12**-

15 and is discussed in further detail below. Most generally, the latch assembly **138** is attached to one of the lid **116** and the base **118** and allows the user to securely fasten the lid **116** to the base **118** by utilizing mechanical advantage. The sterilization container **102** may comprise more than one latch assembly **138** to facilitate thorough and consistent sealing of the volume **120**. In the exemplary sterilization container **102** shown in FIG. **1**, two latch assemblies **138** are attached to the lid **116** at opposing ends **110** of the sterilization container **102**. In other implementations of the sterilization container (not shown), other quantities of latch assemblies may be used to secure the lid and the base. For example, larger containers may utilize more than one latch assembly per side, or square sterilization containers may utilize one latch on each side, for a total of four latch assemblies.

(28) To this end, the latch assembly **138** may comprise a mount body **160**, a lever body **230**, and a clasp body **232**. As will be described in further detail below, the mount body **160** may be fixedly coupled to the lid **116**, the lever body **230** may be coupled to the mount body **160**, and the clasp body **232** may be coupled to the lever body **230**. In some configurations the mount body **160** may be coupled to the base **118** and configured such that the clasp body **232** engages the lid **116** to fasten the base **118** to the lid **116**. The lever body **230** is pivotably coupled to the mount body **160** such that the lever body **230** is moveable between an unsecured position and a secured position. By moving the lever body **230** between the secured position and unsecured position, a user may secure/unsecure the lid **116** to/from the base **118** without needing to separately touch the clasp body **232** (described below). As illustrated herein, figures designated with “A” correspond to the secured position, and figures designated with “B” correspond to the unsecured position (with the exception of FIG. **3**B). Specifically, in FIGS. **2**A, **3**A, **4**A, and **5**A, the latch assembly **138** is shown in the secured position, and in FIGS. **2**B, **3**C, **4**B, and **5**B the latch assembly **138** is shown in the unsecured position. As will be discussed below, in FIG. **3**B the latch assembly **138** is shown in an intermediate position between the secured position and the unsecured position.

(29) Shown best in FIGS. **5**A and **5**B, the base **118** may comprise a lip **148** integrally formed with the base **118**. This is advantageous because, during transfer of the sterilization container **102**, the base **118** may contact a non-sterile surface. More generally stated, when removing sterile contents from the sterilization container **102**, it is advantageous to limit contact between a user and the sterilization container **102** when removing the sterile contents. As such, because the user may remove the lid **116** of the sterilization container **102** from the base **118** of the sterilization container **102** without separately contacting the base **118** and/or the clasp body **232**, the user is able to remove sterile contents from the sterilization container **102** in a sterile manner.

(30) As mentioned above, the mount body **160** may be fixedly coupled to the lid **116**, and as shown in the figures, may be connected to one of the ends **110** of the lid **116**. Here, the lid **116** includes two latch assemblies **138**, which are arranged on the shorter of two pairs of opposing sides. Best shown in FIGS. **4**A and **4**B, the mount body **160** comprises a back portion **162** coupled to the lid **116** and two side portions **164** that extend from the back portion **162** away from the lid **116**. Better shown in FIG. **7**, several features may be defined in the two side portions **164**, a pivot bore **236** is defined in the mount body **160** and extends between each of the side portions **164** and defines a pivot axis **238**. The pivot axis **238** is generally parallel to the side of the lid **116** and configured to receive a pivot shaft **240**, as will be discussed in further detail below. The back portion **162** of the mount body **160** may define a mount body recess **166** sized and shaped to receive a tab **186** for holding the lever body **230** in the unsecured position. The mount body recess **166** shown here is in the form of a circular depression formed in the back portion **162** and facing the lever body **230**.

(31) Operation of the latch assembly **138** is effected via the lever body **230**. The lever body **230** has a handle portion **246** and a body portion **248**, the handle portion **246** is configured to be grasped by a user in furtherance of operating the latch assembly **138** and the body portion **248** is configured to effect coordinated movement of the latch assembly **138** in response to actuation of the handle portion **246**. The body portion **248** of the lever body **230** may comprise a front wall **250** and two

side walls **252**. The side walls **252** extend in a generally perpendicular direction from opposing sides of the front wall **250** toward an edge. The front wall **250** and the side walls **252** may be formed, for example, by bending opposite edges of a flat material to form a U shape. A pair of wings **258** protrude from the front wall **250** in a generally parallel direction to partially form the handle portion **246** of the lever body **230**. As shown in FIG. **7**, a pivot aperture **260** and a link aperture **262** are defined in the body portion **248** of the lever body **230**, each extending through at least one of the side walls **252**. The pivot aperture **260** is configured to receive the pivot shaft **240** and the link aperture **262** is configured to receive a link shaft **244**, also discussed in further detail below. The link aperture **262** is radially spaced from the pivot axis **238** such that, when viewed from a direction parallel with the pivot axis **238**, the link aperture **262** traces an arcuate path as the lever body **230** is moved between the secured position and the unsecured position.

(32) The lever body **230** is disposed on the mount body **160** with the side walls **252** positioned adjacent to the side portions **164** of the mount body **160** such that the pivot aperture **260** in the side walls **252** are aligned with the pivot bore **236** of the mount body **160**. The pivot shaft **240** is inserted through the pivot apertures **260**, thereby pivotably coupling the lever body **230** to the mount body **160**. The mount body **160** and the lever body **230** cooperate to define a latch interior **204** when coupled, and further define an opening **206** of the latch interior **204** when the lever body **230** is in the secured position. The interior **204** defines an interior width **208** (FIG. **7**), which may be measured between each of the side portions **164** of the mount body **160**.

(33) Turning now to FIGS. **3**A-**3**C, the lever body **230** is shown in a secured position (FIG. **3**A), an intermediate position (FIG. **3**B), and an unsecured position (FIG. **3**C). The lever body **230** is pivotable, relative to the mount body **160**, about the pivot axis **238** between the secured position and the unsecured position. The secured position is generally defined by the lever body **230** being arranged approximately parallel to the back portion **162** of the mount body **160**, and the handle portion **246** positioned relatively near the base **118** of the sterilization container **102**. The unsecured position is generally defined by the lever body **230** being arranged approximately perpendicular to the back portion **162** of the mount body **160**, and the handle portion **246** positioned relatively far from the base **118** of the sterilization container **102**. Said differently, the handle portion **246** is positioned closer to the base **118** in the secured position than in the unsecured position. While parallel and perpendicular are used to generally describe the position the lever body **230** with respect to other features of the latch assembly **138**, they are merely terms of description rather than precise measurements of the position of the specific components to which they are referencing. In this way, it is contemplated that in the secured position the front wall **250** of the lever body **230** could be at an angle that is within approximately 30° of parallel to the back portion **162** of the mount body **160**. Likewise, in the unsecured position the front wall **250** of the lever body **230** could be at an angle that is within approximately 30° of perpendicular to the back portion **162** of the mount body **160**.

(34) In addition to being disposed in both the pivot bore **236** and the pivot aperture **260**, the pivot shaft **240** may be secured in position or to either of the lever body **230** and/or the mount body **160** via several methods. For example, one exemplary method may enlarge opposing ends of the pivot shaft **240** to prevent the pivot shaft **240** from sliding out of the pivot apertures **260**. Alternatively, a press fit between the pivot shaft **240** and the pivot bore **236** such that the lever body **230** pivots relative to the pivot shaft **240** may be utilized. A press fit between the pivot shaft **240** and the pivot aperture **260** may be utilized such that the pivot shaft **240** moves with the lever body **230** relative to the mount body **160** may also be utilized. Further methods, such as staking, fasteners, welding, and the like may also be utilized either in the alternative or in combination.

(35) Turning to FIGS. **6** and **7**, movement of the lever body **230** is transferred to the base **118** via the clasp body **232**, which is coupled to the lever body **230**. The clasp body **232** has an interface end **264** and a link end **266**. The interface end **264** is configured to engage the lip **148** of the base **118** for tensioning the lid **116** toward the base **118**. The link end **266** defines a clasp bore **268** and a

link axis **242**, which is configured to receive the link shaft **244** such that the clasp body **232** is coupled to the lever body **230** and movable about the link shaft **244**. Movement of the link end **266** of the clasp body **232** corresponds to movement of the link aperture **262** in the lever body **230**, which moves about a semi-circular arc about the pivot axis **238** as the lever body **230** moves between the secured position and the unsecured position. As shown in FIGS. **6** and **7**, the clasp body **232** may further comprise two clasp portions **234**A, **234**B with a pocket **270** defined therebetween. Said differently, the clasp body **232** comprises a first clasp portion **234**A and a second clasp portion **234**B with a pocket **270** defined therebetween. The clasp portions **234**A, **234**B extend between the interface end **264** and the link end **266** and are spaced so as to receive a portion of the lever body **230** in the pocket **270** as the lever body **230** is moved toward the secured position.

(36) In some configurations, the clasp bore **268** may be formed on the link end **266** of the clasp body **232** by bending an end of each of the clasp portions **234**A, **234**B around and back toward the interface end **264** at a radius suitable to receive the link shaft **244**. The interface end **264** may be similarly bent to form a hooked profile **272** that is suitable to engage the lip **148** of the base **118** such that when the clasp body **232** is engaging the base **118** and the lever body **230** is in the secured position the interface end **264** is not readily disengaged. In other instances, such as instances wherein the interface end **264** does not include the hooked profile **272** and/or the base **118** does not include the lip **148**, the interface end **264** may be configured to engage with the base **118** via alternative means.

(37) In addition to the hooked profile **272**, the clasp body **232** may be formed with an arcuate profile **274** between the interface end **264** and the link end **266**. Said differently, the first and second clasp portions **234**A, **234**B may be curved between the interface end **264** and the link end **266**. A clasp offset R**3** is defined between the clasp body **232** and the pivot axis **238**. The clasp offset R**3** is a measurement of the distance between an outer surface of the clasp body **232** and the pivot axis **238**.

(38) With renewed reference to FIGS. **1-2**B, in some instances, a frangible sealing element (not shown) may be coupled to the latch assembly **138**. The frangible sealing element may be configured to indicate whether the latch assembly **138** is in the unsecured position or the secured position. Here, the mount body **160** may include a flange extending away from the lid **116**. The flange may have a lock portion **276** that defines a security aperture **278**. The lever body **230** may further define a shear aperture **280** arranged on the body portion **248** and extending through the front wall **250**. The shear aperture **280** is arranged such that as the lever body **230** is moved toward the secured position the shear aperture **280** receives the lock portion **276** of the flange and in the unsecured position the shear aperture **280** is spaced from the lock portion **276**. By moving the lever body **230** from the secured position to the unsecured position, the shear aperture **280** of the lever body **230** severs the frangible sealing element. Furthermore, the frangible sealing element may be disposed in the security aperture **278** of the mount body **160**. As such, when the lever body **230** is moved to the secured position and the frangible sealing element is disposed in the security aperture **278**, the frangible sealing element will be severed by the shear aperture **280** when the lever body **230** is moved to the unsecured position.

(39) In various instances, the latch assembly **138** may vary. Additionally, as previously stated, while the base **118** includes a lip **148** and the interface end **264** of the clasp body **232** includes a hooked profile **272**, in other instances the interface end **264** may not include the hooked profile **272** and/or the base **118** may not include the lip **148**. In such instances, the interface end **264** may be configured to engage with the base **118** via alternative means.

(40) As mentioned above, the link shaft **244** is disposed in the link aperture **262** and the clasp bore **268**. Similar to the pivot shaft **240** described above, the link shaft **244** may be secured to the link aperture **262** or the clasp bore **268** by various methods such as, for example, a press fit, welding, fasteners, adhesives, enlarging the ends, and the like. For example, one exemplary method may

enlarge each end of the link shaft **244** using a press, which prevents the link shaft **244** from being removed. Alternatively, a press fit between the link shaft **244** and the clasp bore **268** such that the lever body **230** moves freely on the link shaft **244** may be utilized. A press fit between the link shaft **244** and the link aperture **262** may be utilized such that the clasp body **232** moves freely on the link shaft **244** may also be utilized.

(41) Returning to FIGS. **4**A-**9**, the latch assembly **138** may further comprise a deflector **180** positioned at least partially within the interior **204** defined by the mount body **160** and the lever body **230**. The deflector **180** is movable with the lever body **230** for concurrent movement about the pivot axis **238**. As the lever body **230** is pivoted between the secured position and the unsecured position, the deflector **180** pivots about the pivot axis **238** in a similar manner. The deflector **180** may comprise a hub portion **182**, a wing portion **184**, and a tab **186**. The hub portion **182** is supported on the pivot axis **238** and defines a hub bore **188**. The hub bore **188** is sized and shaped to receive the pivot shaft **240** and aligned with the pivot axis **238**. The hub portion further defines a link bore **202** radially spaced from the hub bore **188**. The link bore **202** is aligned with the link axis **242** and sized and shaped to receive the link shaft **244**. As will be discussed in further detail below, the link bore **202** is positioned at a third angular location **214**. The hub portion **182** has a width that is sized to fit within the pocket **270** between the clasp portions **234**A, **234**B of the clasp body **232**, and as such, the clasp body **232** moves around the deflector **180** as the lever body **230** is moved between the secured position and the unsecured position. The

(42) Best shown in FIGS. **7-9**, the deflector **180** may comprise the wing portion **184**, which may extend from the hub portion **182** and define an outer surface **190** spaced from the pivot axis **238**. The outer surface **190** is positioned adjacent to the opening **206** of the latch interior **204** when the lever body **230** is in the secured position. The wing portion **184** is arranged such that the outer surface **190** prevents access to the latch interior **204** through the opening **206**. As will be appreciated by FIGS. **3**A-**3**C, the opening **206** opens toward the top **112** of the sterilization container **102**, and as the lever body **230** is moved from the secured position (FIG. **3**A), through the intermediate position (FIG. **3**B), and toward the unsecured position (FIG. **3**C), the size of the opening **206** decreases. Due to the location of the opening **206** relative to the lid **116**, the opening **206** presents a potential pinch point for the user during use. More specifically, while opening the sterilization container **102** a user may position one of their fingers in or near the opening **206**, and as lever body **230** is pivoted toward the unsecured position the user's finger(s) may be pinched between the lever body **230** and the lid **116**.

(43) In order to minimize the possibility of the user being pinched by the lever body **230**, the wing portion **184** of the deflector **180** shields the opening **206** of the latch interior **204**, preventing the user's fingers from passing through the opening **206**. In order to block the opening **206**, the wing portion **184** extends from the hub portion **182** in a direction substantially parallel to the pivot axis **238** and defines a deflector width **198**. The wing portion **184** is positioned on the hub portion **182** at a first angular location **210** relative to the pivot axis **238**. Said differently, the wing portion **184** is radially spaced from the pivot axis **238** and defines the first angular location **210**. In FIG. **5**B, the first angular location **210** is illustrated at an upper right position relative to the hub bore **188**. In FIG. **5**A, the first angular location **210** is illustrated at a bottom position relative to the hub bore **188**.

(44) In order to further minimize the possibility of a user being pinched, the wing portion **184** extends from the hub portion **182** such that the opening **206** is blocked so as to prevent a user's fingers, and other objects, from being pinched by the latch assembly **138**. As such, the wing portion **184** may be sized and shaped so as to reduce the open area of the opening **206**. The opening **206** may be reduced such that the measurement of any remaining open area is smaller than a user's finger, substantially eliminating the risk of being pinched. Because it is not necessary to completely block the opening **206**, the wing portion **184** need not extend the entire distance of the interior width **208**. Sufficiently shielding the opening **206** may be accomplished with a deflector width **198**

that is 65% of the interior width **208**. Said differently, a ratio of the deflector width **198** to the interior width **208** may be 0.65. In other implementations the ratio of the deflector width **198** to the interior width **208** may be greater than 0.65, such as 0.75, 0.85, or 0.95.

(45) In addition to the deflector width **198**, the wing portion **184** further defines an arc angle α, shown in FIG. **9**. The arc angle is an angular measurement of the outer surface **190** of the wing portion **184**. Here, the arc angle α defined by the wing portion **184** may be greater than 75°. In the implementation illustrated herein, the arc angle α may be greater than 90°. The arc angle α may be measured from the first angular location **210**. Said differently, the arc angle α may be centered at the first angular location **210**. The deflector **180** further defines a first deflector radius R**1** between the pivot axis **238** and the first angular location **210**. The first deflector radius R**1** is the distance from an outermost point of the wing portion **184** at the first angular location **210** and the pivot axis **238**. When the lever body **230** is in the secured position the first deflector radius R**1** is approximately equal to the clasp offset R**3**.

(46) In the implementation of the deflector **180** shown here, the wing portion **184** may comprise a first wing portion **184**A and a second wing portion **184**B, each extending from the hub portion **182**. The first wing portion **184**A may extend in a first axial direction and the second wing portion **184**B may extend in a second axial direction. Here, the first axial direction and the second axial direction are opposite to each other and parallel to the pivot axis **238**. As shown here, the hub portion **182** is generally positioned in the middle of the interior **204** between the side portions **164** of the mount body **160**, and as such, each of the first wing portion **184**A and the second wing portion **184**B extends approximately from the hub portion **182**, toward the corresponding side portion, approximately the same distance.

(47) In addition to the wing portion **184**, the deflector **180** comprises the tab **186**. The tab **186** is coupled to the hub portion **182** separate from the wing portion **184**. The tab **186** may engage the mount body **160** when the lever body **230** is in the unsecured position for retaining the lever body **230** in the unsecured position. The tab **186** may alternatively engage the lid **116** when the lever body **230** is in the unsecured position for retaining the lever body **230** in the unsecured position. Similarly, the tab **186** may be disengaged from the mount body **160** when the lever body **230** is in the secured position. As such, the tab **186** may limit free movement of the lever body **230** from the unsecured position to the secured position.

(48) As mentioned above, the tab **186** limits free movement of the lever body **230**, which is effected via engagement between the tab **186** and the lever body **230**. To this end, the tab **186** extends between a fixed end **192** coupled to the hub portion **182** and a free end **194** spaced from the hub portion **182**. The free end **194** of the tab **186** may comprise a protrusion **196**, which defines a second angular location **212** (discussed below). The second angular location **212** being spaced from the first angular location **210**. The tab **186** may be formed such that the free end **194** is spaced from the hub portion **182**. When the tab **186** is formed from a flexible material, space between the free end **194** and the hub portion **182** allows the free end **194** to move relative closer to the hub portion **182**. Turning to FIG. **7**, the protrusion **196** is able to engage the mount body recess **166** of the mount body **160** to hold the lever body **230** in the unsecured position. Moving the lever body **230** into, or out of, the unsecured position flexes the tab **186**, adding resistance to the movement.

(49) Shown best in FIGS. **5**A and **9**, the deflector **180** further defines a second deflector radius R**2** between the pivot axis and the protrusion **196**. The second deflector radius R**2** is the distance from an outermost point of the protrusion **196** at the second angular location **212** and the pivot axis **238**. The second deflector radius R**2** is greater than the first deflector radius R**1**. The increased distance to the protrusion **196** means that the free end **194** of the tab **186** must be flexed toward the hub portion **182** when the lever body **230** is moved into or out of the unsecured position. Additionally, the second deflector radius R**2** is greater than the clasp radius R**3**. Said differently, the clasp offset R**3** is less than the second deflector radius R**2**.

(50) Pivoting movement of the lever body **230** can be defined relative to the angular locations on

the deflector **180**. The first angular location **210** is spaced from the second angular location **212** with the third angular location **214** positioned therebetween. As illustrated herein, an angular measurement between the first angular location **210** and the second angular location **212** may be less than 90°. When the lever body **230** is in the secured position, the first angular location **210** is nearer to the base **118** and the lid **116** of the sterilization container **102** than the second angular location **212**. When the lever body **230** is in the unsecured position, the second angular location **212** is nearer to the base **118** and the lid **116** of the sterilization container **102** than the first angular location **210**.

(51) Referring again to the side views shown in FIGS. **3**A-**3**C, where the latch assembly **138** is shown in the secured position, the intermediate position, and the unsecured position along with corresponding movement of the clasp body **232**. Movement of the lever body **230** toward the unsecured position moves the clasp body **232** to disengage the interface end **264** from the lip **148** of the base **118**. As the lever body **230** is pivoted the link shaft **244** moves in a semi-circular arc, such that the link shaft **244** moves from a position generally above the pivot shaft **240** to a position generally below the pivot shaft **240** and the link end **266** of the clasp body **232** moves in a downward direction.

(52) Furthermore, movement of the clasp body **232** can be partly defined by the position of the wing portion **184** of the deflector **180**. The wing portion **184** of the deflector **180** is spaced from the clasp body **232** when the lever body **230** is in the secured position and as the lever body **230** and the deflector **180** pivot about the pivot axis **238** away from the secured position, the wing portion **184** engages the clasp body **232** such that as the lever body **230** is further pivoted toward the unsecured position the wing portion **184** moves the interface end **264** away from the base **118**. In some implementations, the intermediate position of the lever body **230** may be defined at a position where the wing portion **184** contacts the clasp body **232**. At this intermediate position the wing portion **184** engages one of the clasp portions **234**A, **234**B of the clasp body **232** and as the lever body **230** is further pivoted toward the unsecured position the clasp body **232** pivots around the pivot axis **238** and the interface end **264** moves away from the base **118**. Alternatively, in the intermediate position movement of the lever body **230** toward the secured position causes the wing portion **184** to become spaced from the clasp body **232** such that the hooked profile **272** can engage the lip **148** of the base **118**. Similarly, the pivot shaft **240** may protrude from the hub portion **182** such that the pivot shaft **240** is spaced from the clasp body **232** when the lever body **230** is in the secured position. As the lever body **230** is pivoted away from the secured position, the pivot shaft **240** may engage the clasp body **232** such that further pivoting of the lever body **230** toward the unsecured position moved the interface end **264** of the clasp body **232** away from the base **118** and lip **148**.

(53) Attaching and detaching the lid **116** from the base **118** is advantageously performed simultaneously with actuation of the latch assembly **138** because motion of the lever body **230** shares a component direction with the direction that the lid **116** moves relative to the base **118** during attaching and detaching. Owing to the configuration of the latch assembly **138**, movement of the handle portion **246** to engage the lid **116** with the base **118** is continuous with pivoting of the lever body **230** from the unsecured position to the secured position, therefore the lid **116** can be coupled to the base **118** with a single motion. Specifically, with the lever body **230** in the unsecured position a user grasps the handle portion **246** and moves the lid **116** downward to engage the base **118**, upon engagement of the lid and the base **118** the user continues with the downward motion to pivot the lever body **230** from the unsecured position to the secured position, thereby moving the clasp body **232** into engagement with the base **118** and securing the lid **116** to the base **118**.

(54) The latch assembly **138** is configured to effect disengaging the lid **116** from the base **118** in a similarly continuous movement. Pivoting the lever body **230** toward the unsecured position to effect disengagement of the interface end **264** of the clasp body **232** from the lip **148** of the base **118** is continuous with movement of the handle portion **246** to disengage the lid **116** from the base

118. Specifically, with the lever body **230** in the secured position as shown in FIG. **3**A, a user grasps the handle portion **246** and pivots the lever body **230** toward the unsecured position by way of the intermediate position shown in FIG. **3**B, causing the interface end **264** of the clasp body **232** to move downward and disengage from the lip **148**. In the intermediate position, the link end **266** of the clasp body **232** has moved downward such that one of the clasp portions **234**A, **234**B contacts the wing portion **184**. As the user continues to move the lever body **230** toward the unsecured position the handle portion **246** moves upwardly, which causes the link end **266** to correspondingly move downward. Due to the contact between the clasp body **232** and the wing portion **184**, the interface end **264** moves outwardly away from the lip **148**, and upon reaching the unsecured position as shown in FIG. **3**C the user continues with the upward movement to lift the lid **116** away from the base **118**. Due to the contact between the clasp body **232** and the pivot shaft **240** which causes coordinated movement between the lever body **230** and the clasp body **232**, the user is not required to perform a secondary step of disengaging the interface end **264**, and as such can remove and attach the lid **116** to the base **118** by only contacting the handle portion **246** of the lever body **230**.

(55) Turning now to FIGS. **10-13**, another version of a latch assembly and deflector are shown. As will be appreciated from the subsequent description below, the second latch assembly and deflector are similar to the latch assembly **138** and deflector **180** described above in connection with FIGS. **2**A-**9**. As such, the components and structural features of the second version of the latch assembly **138**′ and deflector **180**′ that are the same as, or that otherwise correspond to, the first version of the latch assembly **138** and deflector **180** are provided with the same reference numerals with the addition of a prime symbol (e.g., **138** and **138**′). While the specific differences between these versions will be described in detail, for the purposes of clarity, consistency, and brevity, only certain structural features and components common between these versions will be discussed and depicted in the drawings of the second version of the latch assembly **138**′ and deflector **180**′. Here, unless otherwise indicated, the above description of the first version of the latch assembly **138** and deflector **180** may be incorporated by reference with respect to the second version of the latch assembly **138**′ and deflector **180**′ without limitation.

(56) Similar to above, the latch assembly **138**′ may comprise a mount body **160**′, a lever body **230**′, and a clasp body **232**′. As will be described in further detail below, the mount body **160**′ may be fixedly coupled to the lid, the lever body **230**′ may be coupled to the mount body **160**′, and the clasp body **232**′ may be coupled to the lever body **230**′. In some configurations the mount body **160**′ may be coupled to the base and configured such that the clasp body **232**′ engages the lid to fasten the base to the lid. The lever body **230**′ is pivotably coupled to the mount body **160**′ such that the lever body **230**′ is moveable between an unsecured position and a secured position. By moving the lever body **230**′ between the secured position and unsecured position, a user may secure/unsecure the lid to/from the base without needing to separately touch the clasp body **232**′.

(57) Here too, the mount body **160**′ comprises a back portion **162**′ coupled to the lid and two side portions **164**′ that extend from the back portion **162**′ away from the lid. Several features may be defined in the two side portions **164**′, a pivot bore **236**′ is defined in the mount body **160**′ and extends between each of the side portions **164**′ and defines a pivot axis **238**′. The pivot axis **238**′ is generally parallel to the side of the lid and configured to receive a pivot shaft **240**′. The back portion **162**′ of the mount body **160**′ may define a mount body recess **166**′ sized and shaped to receive a tab **186**′ for holding the lever body **230**′ in the unsecured position. The mount body recess **166**′ shown here is in the form of a circular depression formed in the back portion **162**′ and facing the interior **204**′.

(58) The lever body **230**′ has a handle portion **246**′ and a body portion **248**′, the handle portion **246**′ is configured to be grasped by a user in furtherance of operating the latch assembly **138**′ and the body portion **248**′ is configured to effect coordinated movement of the latch assembly **138**′ in response to actuation of the handle portion **246**′. The body portion **248**′ of the lever body **230**′ may

comprise a front wall **250′** and two side walls **252′**. The side walls **252′** extend in a generally perpendicular direction from opposing sides of the front wall **250′** toward an edge. The front wall **250′** and the side walls **252′** may be formed, for example, by bending opposite edges of a flat material to form a U shape. A pair of wings **258′** protrude from the front wall **250′** in a generally parallel direction to partially form the handle portion **246′** of the lever body **230′**. A pivot aperture **260′** and a link aperture **262′** are defined in the body portion **248′** of the lever body **230′**, each extending through at least one of the side walls **252′**. The pivot aperture **260′** is configured to receive the pivot shaft **240′** and the link aperture **262′** is configured to receive a link shaft **244′**. The link aperture **262′** is radially spaced from the pivot axis **238′** such that, when viewed from a direction parallel with the pivot axis **238′**, the link aperture **262′** traces an arcuate path as the lever body **230′** is moved between the secured position and the unsecured position.
(59) The lever body **230′** is disposed on the mount body **160′** with the side walls **252′** positioned adjacent to the side portions **164′** of the mount body **160′** such that the pivot aperture **260′** in the side walls **252′** are aligned with the pivot bore **236′** of the mount body **160′**. The pivot shaft **240′** is inserted through the pivot apertures **260′**, thereby pivotably coupling the lever body **230′** to the mount body **160′**. The mount body **160′** and the lever body **230′** cooperate to define a latch interior **204′** when coupled, and further define an opening **206′** of the latch interior **204′** when the lever body **230′** is in the secured position.
(60) The latch assembly **138′** may further comprise a deflector **180′** positioned at least partially within the interior **204′** defined by the mount body **160′** and the lever body **230′**. The deflector **180′** is movable with the lever body **230′** for concurrent movement about the pivot axis **238′**. As the lever body **230′** is pivoted between the secured position and the unsecured position, the deflector **180′** pivots about the pivot axis **238′** in a similar manner. The deflector **180′** may comprise a hub portion **182′**, a wing portion **184′**, and a tab **186′**. The hub portion **182** is supported on the pivot axis **238′** and defines a hub bore **188′** sized and shaped to receive the pivot shaft **240′**. The hub portion **182′** has a width that is sized to fit within the pocket **270′** between the clasp portions **234**A′, **234**B′ of the clasp body **232′**, and as such, the clasp body **232′** moves around the deflector **180′** as the lever body **230′** is moved between the secured position and the unsecured position.
(61) In order to minimize the possibility of a user being pinched by the lever body **230′**, the wing portion **184′** of the deflector **180′** shields the opening **206′** of the latch interior **204′**, preventing anything from passing through the opening **206′**. In order to block the opening **206′**, the wing portion **184′** extends from the hub portion **182′** in a direction substantially parallel to the pivot axis **238′**. The wing portion **184′** is positioned on the hub portion **182′** at a first angular location **210′** relative to the pivot axis **238′**. Said differently, the wing portion **184′** is radially spaced from the pivot axis **238′** and defines the first angular location **210′**.
(62) In addition to the wing portion **184′**, the deflector **180′** comprises the tab **186′**. The tab **186′** is coupled to the hub portion **182′** separate from the wing portion **184′**. The tab **186′** may engage the mount body **160′** when the lever body **230′** is in the unsecured position for retaining the lever body **230′** in the unsecured position. The tab **186′** may alternatively engage the lid when the lever body **230′** is in the unsecured position for retaining the lever body **230′** in the unsecured position. Similarly, the tab **186′** may be disengaged from the mount body **160′** when the lever body **230′** is in the secured position. As such, the tab **186′** may limit free movement of the lever body **230′** from the unsecured position to the secured position.
(63) Similar to above, the tab **186′** limits free movement of the lever body **230′**, which is effected via engagement between the tab **186′** and the lever body **230′**. To this end, the tab **186′** may be coupled to the hub portion **182′** and extend toward a second angular location **212′**. The second angular location **212′** being spaced from the first angular location **210′**. The tab **186′** may be formed from a metal and comprise a fixed end **192′** and a free end **194′**, with the fixed end **192′** coupled to the hub portion **182′** and the free end **194′** extending therefrom. The free end **194′** is spaced from the hub portion **182′** such that the free end **194′** is able to move relative to the hub portion **182′**. The

free end **194′** may comprise a protrusion **196′**, which is able to engage the mount body recess **166′** of the mount body **160′** to hold the lever body **230′** in the unsecured position. Moving the lever body **230′** into, or out of, the unsecured position flexes the tab **186′**, adding resistance to the movement.

(64) Several instances have been discussed in the foregoing description. However, the aspects discussed herein are not intended to be exhaustive or limit the disclosure to any particular form. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects without departing from the scope of the disclosure. The terminology that has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the disclosure may be practiced otherwise than as specifically described.

CLAUSES

(65) I. A sterilization container for medical instruments, the sterilization container comprising: a base and a lid configured for engaging the base, the base and the lid collectively defining a volume for receiving medical instruments; a latch attached to one of the base and the lid, the latch comprising: a mount body including a back portion and two side portions, wherein the back portion is coupled to the lid or the base; a lever body including two side portions, wherein the two side portions of the lever body are pivotably coupled to two side portions of the mount body such that the lever body is movable about a pivot axis between a secured position and an unsecured position, and wherein the lever body and the mount body cooperate to define an interior and further define an opening of the interior when the lever body is in the secured position; and a deflector positioned at least partially within the interior and movable with the lever body for concurrent movement about the pivot axis, the deflector comprising: a hub portion supported on the pivot axis; a wing portion extending from the hub portion to define an outer surface, wherein the outer surface is positioned adjacent to the opening of the interior when the lever body is in the secured position for preventing access to the interior through the opening; and a tab coupled to the hub portion and separate from the wing portion, the tab engages the mount body when the lever body is in the unsecured position for retaining the lever body in the unsecured position. II. The sterilization container of clause I, wherein the tab is disengaged from the mount body when the lever body is in the secured position. III. The sterilization container of clause I, wherein the lever body defines a link axis extending through the two side portions, and further comprising a clasp body at least partially disposed in the interior, the clasp body having an interface end selectively engageable with one of the base and the lid, and a link end, wherein the link end is rotatably supported on the link axis. IV. The sterilization container of clause III, wherein the mount body is coupled to the lid and the clasp body is engageable with the base, and wherein the secured position of the lever body is further defined as the lever body being pivoted toward the base and the unsecured position of the lever body is further defined as the lever body being pivoted toward the lid. V. The sterilization container of clause III, wherein the wing portion defines an arc angle greater than 75 degrees. VI. The sterilization container of clause III, wherein the wing portion comprises a first wing portion extending from the hub portion in a first axial direction and a second wing portion extending from the hub portion in a second axial direction. VII. The sterilization container of clause III, wherein the wing portion extends from the hub portion in a direction parallel to the pivot axis and is positioned at a first angular location. VIII. The sterilization container of clause VII, wherein the tab is positioned at a second angular location spaced from the first angular location. IX. The sterilization container of clause VIII, wherein the hub portion defines a first bore and a second bore, the first bore aligned with the pivot axis and the second bore radially spaced therefrom and aligned with the link axis, and wherein the second bore is positioned at a third angular location. X. The sterilization container of clause IX, wherein the tab extends between a fixed end coupled to the hub portion and a free end spaced from the hub portion, and wherein the free end comprises a protrusion defining the second angular location. XI. The sterilization container of clause X,

wherein a first deflector radius is defined between the pivot axis and an outermost point at the first angular location and a second deflector radius is defined between the pivot axis and an outermost point of the protrusion, and wherein the second deflector radius is greater than the first deflector radius. XII. The sterilization container of clause XI, wherein the clasp body defines an arcuate profile between the interface end and the link end, wherein a clasp offset is defined between the clasp body and the pivot axis, and wherein the clasp offset is less than the second deflector radius. XIII The sterilization container of clause XII, wherein the clasp offset is approximately equal to the first deflector radius when the lever body is in the secured position. XIV. The sterilization container of clause XII, wherein a pivot shaft is arranged on the pivot axis and supports the lever body and the deflector for rotation about the pivot axis, wherein the pivot shaft protrudes from the hub portion such that the pivot shaft is spaced from the clasp body when the lever body is in the secured position and as the lever body is pivoted away from the secured position the pivot shaft engages the clasp body such that further pivoting the lever body toward the unsecured position moves the interface end of the clasp body away from the base. XV. The sterilization container of clause IX, wherein when the lever body is in the secured position the first angular location is nearer to the container than the second angular location, and when the lever body is in the unsecured position the second angular location is nearer to the container than the first angular location. XVI. The sterilization container of clause III, wherein the clasp body comprises a first clasp portion and a second clasp portion with a pocket defined therebetween, and wherein the hub portion of the deflector is at least partially arranged in the pocket. XVII. The sterilization container of clause XVI, wherein the wing comprises a first wing portion extending from the hub portion arranged adjacent to the first clasp portion, and a second wing portion extending from the hub portion arranged adjacent to the second clasp portion. XVIII. The sterilization container of clause I, wherein the interior defines an interior width, and the wing portion defines a deflector width, and wherein a ratio of the deflector width to the interior width is at least 0.65.

Claims

1. A sterilization container for medical instruments, the sterilization container comprising: a base and a lid configured for engaging the base, the base and the lid collectively defining a volume for receiving medical instruments; a latch attached to one of the base and the lid, the latch comprising: a mount body including a back portion and two side portions, wherein the back portion is coupled to the lid or the base; a lever body including two side portions, wherein the two side portions of the lever body are pivotably coupled to the two side portions of the mount body such that the lever body is movable about a pivot axis between a secured position and an unsecured position, and wherein the lever body and the mount body cooperate to define an interior and further define an opening of the interior when the lever body is in the secured position; and a deflector positioned at least partially within the interior and movable with the lever body for concurrent movement about the pivot axis, the deflector comprising: a hub portion supported on the pivot axis; a wing portion extending from the hub portion to define an outer surface, wherein the outer surface is positioned adjacent to the opening of the interior when the lever body is in the secured position for preventing access to the interior through the opening; and a tab coupled to the hub portion and separate from the wing portion, wherein the tab is arranged to engage the mount body when the lever body is in the unsecured position for retaining the lever body in the unsecured position; wherein the lever body defines a link axis extending through the two side portions, and further comprising a clasp body at least partially disposed in the interior, the clasp body having an interface end selectively engageable with the other one of the base and the lid, and a link end, wherein the link end is rotatably supported on the link axis; and wherein the wing portion comprises a first wing portion extending from the hub portion in a first axial direction and a second wing portion extending from the hub portion in a second axial direction, wherein the first wing portion and the second wing

portion are configured to engage the clasp body and move to disengage the interface.
2. The sterilization container of claim 1, wherein the tab is disengaged from the mount body when the lever body is in the secured position.
3. The sterilization container of claim 1, wherein the mount body is coupled to the lid and the clasp body is engageable with the base, and wherein the secured position of the lever body is further defined as the lever body being pivoted toward the base and the unsecured position of the lever body is further defined as the lever body being pivoted toward the lid.
4. The sterilization container of claim 1, wherein the wing portion defines an arc angle greater than 75 degrees.
5. The sterilization container of claim 1, wherein the wing portion extends from the hub portion in a direction parallel to the pivot axis and is positioned at a first angular location.
6. The sterilization container of claim 5, wherein the tab is positioned at a second angular location spaced from the first angular location.
7. The sterilization container of claim 6, wherein the hub portion defines a first bore and a second bore, the first bore aligned with the pivot axis and the second bore radially spaced therefrom and aligned with the link axis, and wherein the second bore is positioned at a third angular location.
8. The sterilization container of claim 7, wherein the tab extends between a fixed end coupled to the hub portion and a free end spaced from the hub portion, and wherein the free end comprises a protrusion defining the second angular location.
9. The sterilization container of claim 8, wherein a first deflector radius is defined between the pivot axis and an outermost point at the first angular location and a second deflector radius is defined between the pivot axis and an outermost point of the protrusion, and wherein the second deflector radius is greater than the first deflector radius.
10. The sterilization container of claim 9, wherein the clasp body defines an arcuate profile between the interface end and the link end, wherein a clasp offset is defined between the clasp body and the pivot axis, and wherein the clasp offset is less than the second deflector radius.
11. The sterilization container of claim 10, wherein the clasp offset is approximately equal to the first deflector radius when the lever body is in the secured position.
12. The sterilization container of claim 10, wherein a pivot shaft is arranged on the pivot axis and supports the lever body and the deflector for rotation about the pivot axis, wherein the pivot shaft protrudes from the hub portion such that the pivot shaft is spaced from the clasp body when the lever body is in the secured position and as the lever body is pivoted away from the secured position the pivot shaft engages the clasp body such that further pivoting the lever body toward the unsecured position moves the interface end of the clasp body away from the base.
13. The sterilization container of claim 7, wherein when the lever body is in the secured position the first angular location is nearer to the lid than the second angular location, and when the lever body is in the unsecured position the second angular location is nearer to the lid than the first angular location.
14. The sterilization container of claim 1, wherein the clasp body comprises a first clasp portion and a second clasp portion with a pocket defined therebetween, and wherein the hub portion of the deflector is at least partially arranged in the pocket.
15. The sterilization container of claim 14, wherein the first wing portion extends from the hub portion and is arranged adjacent to the first clasp portion, and the second wing portion extends from the hub portion and is arranged adjacent to the second clasp portion.
16. The sterilization container of claim 1, wherein the interior defines an interior width, and the wing portion defines a deflector width, and wherein a ratio of the deflector width to the interior width is at least 0.65.
17. A latch for a sterilization container for medical instruments having a base and a lid removably engageable with the base, the latch comprising: a mount body coupled to one of the lid and the base and including two side portions; a lever body including two side portions, wherein each of the two

side portions of the lever body are pivotably coupled to one of the two side portions of the mount body such that the lever body is movable about a pivot axis between a secured position and an unsecured position, and wherein the lever body and the mount body cooperate to define an interior and further define an opening of the interior when the lever body is in the secured position; and a deflector supported on the pivot axis and positioned at least partially within the interior and movable with the lever body about the pivot axis, the deflector comprising: a hub portion; a wing portion extending from the hub portion to define an outer surface, wherein the outer surface is positioned adjacent to the opening of the interior for preventing access to the interior through the opening; and a tab coupled to the hub portion and arranged to engage the mount body when the lever body is in the unsecured position for retaining the lever body in the unsecured position; wherein the lever body defines a link axis extending through the two side portions, and further comprising a clasp body at least partially disposed in the interior, the clasp body having an interface end selectively engageable with the other one of the base and the lid, and a link end, wherein the link end is rotatably supported on the link axis; and wherein the wing portion comprises a first wing portion extending from the hub portion in a first axial direction and a second wing portion extending from the hub portion in a second axial direction, wherein the first wing portion and the second wing portion are configured to engage the clasp body and move to disengage the interface.

18. A sterilization container for medical instruments, the sterilization container comprising: a base and a lid configured for engaging the base, the base and the lid collectively defining a volume for receiving medical instruments; a latch attached to one of the base and the lid, the latch comprising: a mount body including a back portion and two side portions, wherein the back portion is coupled to the lid or the base; a lever body including two side portions, wherein the two side portions of the lever body are pivotably coupled to the two side portions of the mount body such that the lever body is movable about a pivot axis between a secured position and an unsecured position, and wherein the lever body and the mount body cooperate to define an interior and further define an opening of the interior when the lever body is in the secured position; and a deflector positioned at least partially within the interior and movable with the lever body for concurrent movement about the pivot axis, the deflector comprising: a hub portion supported on the pivot axis; a wing portion extending from the hub portion to define an outer surface, wherein the outer surface is positioned adjacent to the opening of the interior when the lever body is in the secured position for preventing access to the interior through the opening; and a tab coupled to the hub portion and separate from the wing portion, wherein the tab is arranged to engage the mount body when the lever body is in the unsecured position for retaining the lever body in the unsecured position; wherein the lever body defines a link axis extending through the two side portions, and further comprising a clasp body at least partially disposed in the interior, the clasp body having an interface end selectively engageable with the other one of the base and the lid, and a link end, wherein the link end is rotatably supported on the link axis; and wherein the wing portion extends from the hub portion in a direction parallel to the pivot axis and is positioned at a first angular location.